DATABASE DESIGN ANALYSIS

Assignment 3

DATABASE MANAGEMENT SYSTEM

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Introduction

Database design analysis is an important process in developing an efficient and reliable database system. It helps identify data redundancy, insertion anomalies, deletion anomalies, and update anomalies. It also involves identifying functional dependencies between attributes and finding suitable ways to improve the database structure. In this assignment, the Food Delivery Management System is analysed to identify potential database design problems and propose improvements to maintain data accuracy and consistency.

Data Redundancy and Potential Anomalies

Data Redundancy:

- Data redundancy occurs when the same information is stored repeatedly in a database. It increases storage requirements and may lead to inconsistent data.
- In the Food Delivery Management System, redundancy may occur when customer details, restaurant information, or delivery partner contact details are repeated across multiple records.
- For example, if a delivery partner's name and phone number are stored repeatedly for different orders, updating the phone number may require changes to multiple records.
- The existing database reduces redundancy by storing customer, restaurant, menu item, order, payment, and delivery information in separate tables.

Insertion Anomaly:

- An insertion anomaly occurs when new information cannot be added without inserting unrelated information.
- **Example:** If restaurant details and menu items were stored in the same table, a new restaurant might not be added until at least one menu item was available.
- **Improvement:** Store restaurant information in the RESTAURANT table and menu items in the MENU_ITEM table.

Deletion Anomaly:

- A deletion anomaly occurs when deleting one record unintentionally removes other important information.
- Example: If restaurant details and menu items were stored together, deleting the last menu item could also remove the restaurant's details.

- Improvement: Maintain separate RESTAURANT and MENU_ITEM tables so that menu items can be deleted independently.

Update Anomaly:

- An update anomaly occurs when the same information is stored in multiple records and all occurrences must be updated.
- Example: If a delivery partner's phone number is repeated across several records, changing the number requires updating every occurrence. Missing a record may result in inconsistent information.
- Improvement: Store delivery partner details separately and identify each partner using a unique identifier.

Functional Dependencies Between Attributes

A functional dependency describes a relationship in which one attribute determines another attribute. It is represented as X → Y, meaning that X uniquely determines Y.

The functional dependencies in the Food Delivery Management System are:

Table	Functional Dependencies
CUSTOMER	Customer_ID , Customer_Name, Phone, Email, Address
RESTAURANT	Restaurant_ID ,Restaurant_Name, Phone, Rating, Address
MENU_ITEM	Item_ID ,Restaurant_ID, Item_Name, Category, Price, Availability
ORDERS	Order_ID ,Customer_ID, Restaurant_ID, Order_Date, Total_Amount, Order_Status
ORDER_ITEM	Order_Item_ID , Order_ID, Item_ID, Quantity, Subtotal
PAYMENT	Payment_ID → Order_ID, Payment_Date, Payment_Method, Amount, Payment_Status
DELIVERY_PARTNER	Delivery_ID → Order_ID, Partner_Name, Phone, Delivery_Status

In these dependencies, each primary key uniquely identifies a record and determines the other attributes in its table.

How Poor Database Design Can Lead to Anomalies

Poor database design can cause data redundancy, inconsistency, and difficulties when inserting, updating, or deleting records.

- **Insertion anomaly:** New information may not be added independently if unrelated information is stored in the same table.
- **Deletion anomaly:** Deleting one record may unintentionally remove other important information.
- **Update anomaly:** Repeated information may need to be updated in multiple places, increasing the possibility of inconsistent data.

For example, storing customer details repeatedly in order records can result in different phone numbers being recorded for the same customer. Similarly, combining restaurant and menu information can make it difficult to delete a menu item without affecting restaurant details.

Therefore, poor database design reduces data accuracy, increases maintenance effort, and affects the reliability of the Food Delivery Management System.

Proposed Improvements to the Database Design

The following improvements can make the Food Delivery Management System more efficient and reliable.

1. **Separate related information:** Store customer, restaurant, menu item, order, order item, payment, and delivery partner information in their respective tables to reduce redundancy.
2. **Improve delivery partner storage:** Store delivery partner details using a unique Partner_ID, along with Partner_Name and Phone. Reference Partner_ID from ORDERS to avoid repeating partner contact information for every order.
3. **Use primary and foreign keys:** Assign primary keys to identify records uniquely and foreign keys to establish relationships between tables and maintain referential integrity.
4. **Apply appropriate constraints:** Use UNIQUE constraints for values that must not be duplicated, CHECK constraints for valid prices and quantities, and DEFAULT constraints for appropriate initial values such as an order status of 'Pending'.
5. **Maintain consistent order amounts:** Calculate item subtotals using quantity and the applicable unit price. Ensure that the order total agrees with the items and any applicable charges or discounts.
6. **Avoid unnecessary duplication:** Store each piece of information in the appropriate table and reference it through keys instead of repeating it across multiple records.

These improvements help reduce data redundancy, prevent anomalies, and maintain the accuracy and consistency of the database.

Conclusion

The analysis of the Food Delivery Management System helps identify potential data redundancy and insertion, deletion, and update anomalies. Understanding functional dependencies helps ensure that attributes are correctly related to their respective tables. By separating related information, applying appropriate keys and constraints, and avoiding unnecessary duplication, the database can become more efficient, consistent, and reliable.